# CascadeRoute-VL: Taming the Reasoning Tax in Vision-Language Models via Online Contextual Bandit Routing

Anonymous CVPR submission

## Abstract

Deploying Vision-Language Models (VLMs) at scale faces a fundamental accuracy-efficiency trade-off: compact models are fast but struggle on complex multi-step reasoning, whereas defaulting to massive frontier architectures or unconstrained test-time reasoning incurs a Multimodal Reasoning Tax—inflating token consumption and latency while degrading performance on perception-dominant queries due to linguistic drift. Analyzing 30,540 multimodal queries across 14 benchmarks and 17 VLMs, we observe a +17.24% accuracy gap between static 72B frontier deployment (78.21% full-set, 77.29% on held-out Test $N$=6,108) and a hindsight oracle selector (95.45%), with compact ($\leq$ 8B) models resolving 63.2% of 72B failures (and 57.6% on strictly non-MCQ open-ended tasks where random chance is near zero). To exploit this complementarity online without domain labels or brittle retraining, we introduce CascadeRoute-VL, an online contextual bandit routing framework. CascadeRoute-VL unites three key mechanisms: (i) Disjoint LinUCB over bounded 896-d multimodal representations with an offline warm-start prior ($w_0$=0.20) and recursive $O(d^2)$ Sherman-Morrison updates; (ii) a deterministic horizon-reserve safety guard (Proposition 1: $C_T \leq B_{\max} + c_{\max}$, achieving 0/6 budget violations vs. 5/6 for Lagrangian Bandits-with-Knapsacks) with budget-aware exploration annealing ($\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$); and (iii) an optional calibrated deferral gate. Across held-out test queries ($N$=6,108), CascadeRoute-VL achieves 78.59% accuracy (+1.30% over static 72B, $p_{\text{Holm}} < 0.05$) while reducing inference spend by 38.7%. Under severe distribution shift ($N_{\text{OOD}}$=17,601), CascadeRoute-VL continuously adapts online to reach 81.59% (+4.79% over periodic retraining, +10.14% over online SGD) while cutting USD cost by 32.0%–44.3% vs. monolithic 72B, retaining robustness under sparse (25%) and noisy (20%) feedback.

## 1. Introduction

Deploying Vision-Language Models (VLMs) in production requires navigating a highly non-uniform cost-accuracy landscape across heterogeneous visual domains [5, 9, 16]. Defaulting every incoming query to monolithic frontier models (72B–78B open-weights checkpoints or proprietary APIs) or unconstrained test-time reasoning exacerbates compute expenditure while paradoxically harming fidelity on perception-heavy queries. Across the full VL-RouterBench [16] matrix ($N$=30,540, 14 benchmarks $\times$ 17 VLMs), Qwen2.5-VL-72B attains 78.21% overall accuracy (77.29% on held-out Test $N$=6,108; 69.24% on General+STEM In-Dist Test $N$=2,523) and fails on 6,655 full-set queries. While a raw 17-arm hindsight oracle reaches 95.45% (+17.24% full-set gap), an independent-chance union control yields 99.99%, reminding us that raw multi-arm oracles inflate under many arms; genuine complementarity instead manifests in specific failure rescues—strictly dense $\leq$ 8B models solve 63.2% (4,209/6,655 all tasks; 57.6% or 2,962/5,138 strictly non-MCQ open-ended queries) of 72B failures, rising to 68.5% (all) / 63.6% (non-MCQ) when adding 3B-active MoE Kimi-VL-A3B (16B total).

Figure 1 exposes two opposing facets of this Multimodal Reasoning Tax. On STEM geometry and algorithmic counting problems (e.g., MathVerse, MathVision, MathVista [22, 38]), explicit reasoning yields decisive accuracy gains: 3B-active/16B-total Kimi-VL-A3B-Thinking [11] reaches 79.6% on MathVista and 47.0% on MathVision, beating 72B Qwen2.5-VL (73.9% and 38.8%). Conversely, on fine-grained optical character recognition and visual grounding (e.g., OCRBench, ChartQA, DocVQA, HallusionBench [14, 20, 23, 24]), unconstrained chain-of-thought reasoning can amplify linguistic hallucination drift [29]: compact Qianfan-VL-8B [10] delivers 86.5% on ChartQA and 87.4% on DocVQA using just 1.2–2.7 output tokens, while unconstrained reasoning models emit hundreds of tokens and risk extraction errors.

Existing routing solutions struggle to harness this

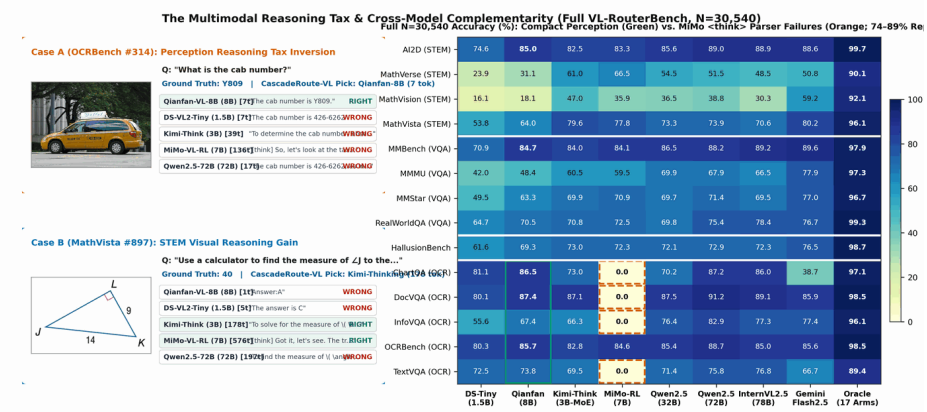



Figure 1. The Multimodal Reasoning Tax and Cross-Model Complementarity (Full VL-RouterBench, $N$=30,540). Left (Top): On OCRBench #314, dense Qianfan-VL-8B correctly reads "Y809" in 7 tokens while Qwen2.5-72B (17 tok), Kimi-Think (39 tok), and MiMo-VL-RL (136 tok) over-rationalize or hallucinate. Left (Bottom): On MathVista #897, compact perception models guess wrong and Qwen2.5-72B confuses $\cos J$ with $\sin J$, whereas reasoning specialist Kimi-VL-A3B-Thinking solves the trigonometry in 178 tokens. Right: Across 14 benchmarks, compact perception excels on Charts/OCR (green box) while unstripped <think> chains cause upstream 0.0% extraction failures on MiMo-VL-7B-RL (dashed orange; repaired to 74.2%–88.7% once <think> is stripped).

complementarity safely under real deployment dynamics:

1. Static Offline Fragility: Multi-label supervised routers in VL-RouterBench [16] and capability-profile classifiers such as ARMS [32] assume stationary query distributions. In our cross-domain stress test (General_VQA + STEM → unseen Charts_OCR + Hallucination, $N_{\text{OOD}}$=17,601), a strong neural MLP-Router collapses from 72.77% to 56.97% (-15.80%) and cannot adapt under gradual drift, price shocks, or test-time model onboarding.
2. Cold-Start Penalty & Heuristic Feature Fragility: Naive online bandits suffer severe initial exploration regret over $K$=17 arms (e.g., Cold LinUCB loses 5.91% accuracy and costs +101% more USD than our warm-start design), while adding handcrafted perceptual cues helps cold start marginally yet significantly degrades cross-domain transfer (−2.41%, $p$=0.0002).
3. Domain-Label Dependence & Budget Overruns: Conditioning on oracle task/domain tags requires brittle classification heads, while unconstrained exploration or soft Lagrange pacing violates tight session USD budgets.

We resolve these limitations with CascadeRoute-VL, an online contextual bandit routing framework uniting three core contributions:

- Domain-Label-Free Multimodal State & Prior Decay: Fuses MiniLM-L6 text [34] and dual-scale DINOv2-ViT-S/14 [28] + orthogonally projected ResNet-50 [15] embeddings with bounded norm ($\|s_t\|_2 \leq 0.7433 < 1$; 896-d default). Offline warm-start regularization ($w_0$=0.20) eliminates cold-start penalties, while online Sherman-Morrison $O(d^2)$ updates guarantee rapid $\mathcal{O}(w_0/(\lambda + n_{t,a}))$ stale prior decay under shift.
- Deterministic Budget Guardrail & Theoretical Regret: Introduces a deterministic horizon reserve (Proposition 1: $C_T \leq B_{\max} + c_{\max}$, achieving 0/6 budget violations vs. 5/6 for Lagrangian Bandits-with-Knapsacks) paired with budget-aware exploration annealing ($\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$) with explicit finite-time elliptical potential bounds (Theorem 1).
- Comprehensive Evaluation Across 30,540 Queries, Shift, & Edge Hardware: Evaluated across 14 benchmarks, 17 VLMs, and external AMBER [33]. Delivers +1.30% accuracy over static 72B on held-out test ($N$=6,108) at 38.7% lower USD cost, outperforms all online adaptive neural baselines under distribution shift (+4.79% over periodic retraining, +10.14% over online SGD), and remains robust under sparse (25%) and noisy (20%) feedback.

## 2. Related Work

LLM and VLM Routing. Early routing work for language models demonstrated substantial inference savings through cascading (FrugalGPT [6]), calibrated binary preference routing (RouteLLM [27]), dual-contrastive selection (RouterDC [8]), and reward-guided ranking (Zooter [21]). Extending routing to multimodal models, VL-RouterBench [16] standardized evaluation over 14 vision-language benchmarks and benchmarked static supervised KNN, linear, and MLP routers. Subsequent methods incorporate static

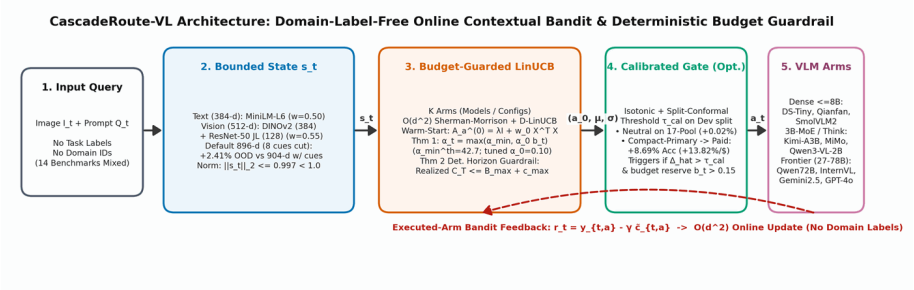



Figure 2. CascadeRoute-VL Architecture. Input image-question pairs are encoded into a domain-label-free bounded context vector $s_t$ ($\|s_t\|_2 \leq 0.997 < 1$; default 896-d Text+Vision after cutting 8 OOD-fragile cues, vs. 904-d ablation) fusing MiniLM-L6 text and dual-scale DINOv2+ResNet-50 vision embeddings. Warm-started Disjoint/Discounted LinUCB with deterministic horizon-reserve budget control ($C_T \leq B_{\max} + c_{\max}$) scores arms, while an optional calibrated deferral gate (*Dev* isotonic/conformal $\tau_{\text{cal}}$) safeguards compact-primary tiers before executed-arm $O(d^2)$ Sherman-Morrison updates adapt online.

model capability profiles (ARMS [32]) or episodic kNN retrieval memory (MHR-Router [41]). Unlike static supervised routers that silently fail under domain shift, or heuristic retrieval stores lacking regret guarantees, CascadeRoute-VL pairs offline warm-start priors with theoretically grounded online contextual bandit adaptation and confidence-gated escalation.

Multimodal Overthinking and Perceptual Grounding. Recent analyses show that excessive test-time reasoning in VLMs can degrade accuracy by amplifying linguistic priors over grounded visual evidence [29]. Our large-scale matrix study across 17 VLMs—including explicit thinking architectures (Kimi-VL-A3B-Thinking [11], MiMo-VL-7B-RL [36]) and perception-specialized compact models (Qianfan-VL-8B [10], DeepSeek-VL2-Tiny [35])—quantifies this dichotomy across reasoning-intensive STEM [17, 22, 38] and OCR/hallucination [14, 20, 30] benchmarks. Rather than modifying VLM weights, CascadeRoute-VL mitigates the Reasoning Tax upfront at routing time.

Contextual Bandits and Budgeted Learning. Disjoint LinUCB [18] and confidence-ellipsoid linear bandits [1] provide finite-time regret guarantees compared to non-contextual bandits [4] and deep heuristic policies [3, 26], while neural variants (NeuralUCB [40], NeuralTS [39]) trade closed-form speed for nonlinear representations. For non-stationary streams, exponentially discounted and sliding-window linear bandits [13] track shifting rewards, and Bandits-with-Knapsacks (BwK) [2] address global budget constraints in expectation via primal-dual pacing. To validate policies without selection bias when only logged bandit data exist, IPS, SNIPS [31], and Doubly Robust (DR) [12] estimators provide counterfactual confidence intervals (reported in Supplementary Sec. S4).

## 3. Quantifying the Multimodal Reasoning Tax

We analyze the complete outcome tensor $Y \in \{0,1\}^{N \times K}$ over $N = 30{,}540$ standardized queries across 14 benchmarks spanning four domains—STEM (AI2D, MathVista, MathVerse, MathVision; $N$=8,332), General_VQA (MMBench, MMMU, MMStar, RealWorldQA; $N$=4,607 [7, 19, 37]), Hallucination (HallusionBench; $N$=951), and Charts_OCR (ChartQA, DocVQA, InfoVQA, OCRBench, TextVQA; $N$=16,650 [25, 30])—paired with $K = 17$ VLMs from 1B to 78B parameters and proprietary APIs (GPT-4o, GeminiFlash2.5). Per-query costs and latencies follow standardized VL-RouterBench pricing and cl100k_base tokenization (+256 visual tokens per image; profiles in Supplementary Sec. S2).

Four structural properties motivate online multimodal routing:

1. Failure Rescues & Chance-Corrected Oracle Check: On the full set ($N$=30,540), Qwen2.5-VL-72B achieves 78.21% (Test $N$=6,108: 77.29%; In-Dist Test $N$=2,523: 69.24%) and fails on 6,655 queries. While a hindsight 17-arm oracle reaches 95.45% (+17.24% raw gap), an independent-chance control yields 99.99%, showing that raw multi-arm oracles inflate by chance under 17 arms. Genuine complementarity lies in specific failure rescues: strictly dense $\leq$ 8B models solve 63.2% (4,209/6,655 all; 57.6% or 2,962/5,138 strictly non-MCQ open-ended queries where chance guessing is $\approx 0\%$) of 72B failures, rising to 68.5% all / 63.6% non-MCQ with 3B-MoE Kimi-VL-A3B. 'Qianfan-8B' alone rescues 38.2% of 72B failures overall (and up to 72.6% on OCRBench).
2. Reasoning Inversion & Data Hygiene Note: On Charts_OCR, concise dense models dominate: Qianfan-VL-8B attains 86.48% on ChartQA (1.2

out tok), 87.42% on DocVQA (2.7 tok), and 85.70% on OCRBench (4.6 tok) at $\sim 5\times$ lower cost than 72B. Crucially, upstream VL-RouterBench logs registered 0.0% for MiMo-VL-7B-RL across four OCR tasks due to an unstripped <think> regex parser bug in VLMEvalKit. Once repaired, MiMo reaches 74.2%–88.7% (DocVQA 87.1%, InfoVQA 66.3

3. Same-Backbone Verification: To eliminate data confounding, our live GPU study on Qwen3-VL-2B (Sec. 5.5) isolates the Reasoning Tax on a single architecture.
4. Domain-Label-Free Foundation Embeddings ($d$=896): Frozen text+vision embeddings (896-d) capture routing structure without task/dataset IDs; retaining the 8 handcrafted cues ($d$=904) is evaluated as an explicit ablation (Sec. 5.4), showing they hurt OOD transfer ($-2.41\%$).

# 4. CascadeRoute-VL Framework

Figure 2 illustrates the complete routing loop. At discrete step $t \in \{1, \ldots, T\}$, an unlabelled multimodal query $(I_t, Q_t)$ arrives without task or dataset identifiers. The router chooses a VLM arm or configuration $a_t \in \mathcal{A} = \{1, \ldots, K\}$, executes inference to observe executed-arm bandit feedback $(y_{t,a_t}, c_{t,a_t}, o_{t,a_t})$, and updates only the executed arm's posterior online (except under dual-arm escalation, where both queried arms are explicitly observed and billed).

## 4.1. Domain-Label-Free Multimodal Context State

To satisfy norm boundedness for linear bandit concentration without leaking benchmark/domain metadata, we construct $s_t \in \mathbb{R}^d$ purely from the raw input pair $(I_t, Q_t)$:

$$s_t = [\, w_q \hat{q}_t \parallel w_v \hat{v}_t \,] \in \mathbb{R}^{896}, \quad \text{or} \quad \left[\, s_t^{(896)} \parallel w_p p_t \,\right] \in \mathbb{R}^{904}, \tag{1}$$

where $(w_q, w_v, w_p) = (0.50, 0.55, 0.235)$ combine:

- Linguistic Semantics ($\hat{q}_t \in \mathbb{R}^{384}$): $L_2$-normalized all-MiniLM-L6-v2 embedding [34].
- Dual-Scale Vision ($\hat{v}_t \in \mathbb{R}^{512}$): $L_2$-normalized concatenation of frozen DINOv2-ViT-S/14 [CLS] (384-d) [28] and a fixed orthonormal Johnson-Lindenstrauss projection ($2048 \to 128$) of ResNet-50 GAP [15].
- Optional Unsupervised Cues ($p_t \in [0,1]^8$): Luminance Shannon entropy $H(I_t)/8$, Sobel edge density, log aspect ratio, log area, white-background ratio ($L > 235/255$), word count, symbolic/MCQ density, and sentence count.

Every coordinate lies strictly inside a unit sphere without task labels ($\|s_t\|_2 \le 0.7433$ for $d$=896, and $\le \sqrt{0.9943} < 1.0$ for $d$=904). As Sec. 5.4 proves, dropping $p_t$ (896-d default) boosts OOD accuracy by $+2.41\%$ ($p$=0.0002). What remains outside input leakage avoidance is the standard warm-start assumption: offline training priors use historical reward logs from seen domains, which decay naturally under $w_0$ when shifted.

## 4.2. Warm-Started Disjoint LinUCB & Cost Reward

We define the cost-adjusted scalar reward for selecting arm $a$ on query $t$ with cost sensitivity parameter $\gamma \ge 0$ as:

$$r_{t,a} = y_{t,a} - \gamma \tilde{c}_{t,a}, \qquad \tilde{c}_{t,a} = \text{clip}\left(\frac{c_{t,a}}{c_{90}}, 0, 2.5\right), \tag{2}$$

where $c_{90}$ = \$0.000329 is the 90th percentile training-arm mean cost, keeping normalized costs invariant across heterogeneous currency scales. Modeling $\mathbb{E}[r_{t,a} \mid s_t] = \theta_a^\top s_t$, we maintain per-arm inverse regularized design matrices $A_a^{-1} \in \mathbb{R}^{d\times d}$ and response vectors $b_a \in \mathbb{R}^d$, yielding ridge parameters $\hat{\theta}_a = A_a^{-1} b_a$ and epistemic uncertainty $\sigma_a(s_t) = \sqrt{s_t^\top A_a^{-1} s_t}$.

Offline Warm-Start Prior. Initializing $A_a = \lambda I_d$ online incurs heavy early exploration regret over $K$=17 VLMs. Given offline training interactions $\mathcal{D}_{\text{off}} = \{(s_i, a_i, r_i)\}_{i=1}^{N_{\text{off}}}$ scaled by prior weight $w_0 \in (0, 1]$ to preserve plasticity under non-stationarity, we initialize:

$$A_a^{(0)} = \lambda I_d + w_0 \sum_{i:\, a_i = a} s_i s_i^\top, \qquad b_a^{(0)} = w_0 \sum_{i:\, a_i = a} r_i s_i. \tag{3}$$

Non-Stationary Tracking & Hot Onboarding. For drifting streams, we pair $O(d^2)$ Sherman-Morrison updates with Discounted LinUCB ($\beta$=0.996: $A_a \leftarrow \beta A_a + (1-\beta)\lambda I_d + s_t s_t^\top$) and Sliding-Window LinUCB ($W$=400). At test time, newly registered models are hot-onboarded online via hot_onboard_arm without retraining offline parameters.

## 4.3. Budget Guardrail and Calibrated Deferral Gate

Given total session budget $B_{\max}$ and cumulative spend $C_{t-1}$, remaining budget fraction is $b_t = \max(0, \min(1, (B_{\max} - C_{t-1})/B_{\max}))$. We modulate exploration with floor-clamped Budget-Aware Exploration Annealing and a Deterministic Horizon Reserve:

$$\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t), \quad a_0 = \arg\max_{a \in \mathcal{A}_t^{\text{feas}}} \left(\hat{\theta}_a^\top s_t + \alpha_t \sigma_a(s_t)\right), \tag{4}$$

where $\mathcal{A}_t^{\text{feas}}$ retains only arms leaving sufficient reserve $(T-t)c_{\min}^{\text{safe}}$ (with cheapest-arm cost bound $c_{\min}^{\text{safe}}$, forcing

Table 1. Held-Out Test Benchmark & Matched-Cost Pareto AUC ($N$=6,108, 17 VLMs, 5 Seeds 42–46, Mean ± Std). CascadeRoute-VL ($\gamma$=0.08) beats static 72B (+1.30%, 95% CI [+0.23, +2.14], Holm-adjusted $p_{\text{Holm}} < 0.05$) with 38.7% USD savings, beats NeuralUCB/NeuralTS by +2.62%/+3.57%, and achieves 78.15% Cost-AUC.

| Algorithm | Category | Test Raw (%) | 5-Split CV (%) | Lat (s) | Cost / 1k ($) | Cost-AUC / Lat-AUC | CR Gain Δ (%) |
|---|---|---|---|---|---|---|---|
| InternVL2.5-78B [9] | Static Frontier | 76.63 | 76.63 ± 0.48 | 1.348 | $0.3020 | 73.20 / 72.85 | +1.96%* |
| Qwen2.5-VL-72B [5] | Static Frontier | 77.29 | 78.21 ± 0.53 | 1.959 | $0.2605 | 74.54 / 74.10 | +1.30%* |
| Per-Domain Best Static | Oracle Domain | 78.11 | — | 1.088 | $0.1868 | — | +0.48% |
| Kimi-VL-3B-Think | 3B-MoE (16B) | 72.86 | — | 0.741 | $0.0699 | (Static Pool) | +5.73%* |
| Qianfan-VL-8B | Dense ≤8B | 69.88 | — | 0.228 | $0.0511 | (Static Pool) | +8.71%* |
| Cost-Matched Random Mix | Baseline | 73.10 | — | 0.985 | $0.1598 | — | +5.49%* |
| Offline-Linear | Offline | 78.21 | — | 1.721 | $0.1777 | 77.24 / 74.01 | +0.38% |
| KNN-Reward (cost-aware) | Offline | 78.19 | — | 1.242 | $0.1134 | 78.81 / 77.37 | +0.40% |
| ARMS-Router [32] | Offline | 78.80 | — | 1.306 | $0.2190 | 77.27 / 76.38 | −0.21% |
| MLP-Router (BCE) [16] | Offline | 79.81 | 80.61 ± 0.66 | 1.845 | $0.1858 | 78.66 / 75.54 | −1.22% |
| MLP-Reward (MSE) | Offline | 79.65 | — | 1.806 | $0.1790 | 78.89 / 76.75 | −1.06% |
| FrugalGPT-Cascade (Calibrated) | Cascade | 76.45 | — | 1.620 | $0.1610 | — | +2.14%* |
| LinUCB (Cold Online) | Online Bandit | 72.68 ± 1.71 | — | 1.535 | $0.3220 | — | +5.91%* |
| MHR-Router [41] | Retr. Bandit | 71.48 ± 0.51 | — | 1.333 | $0.1247 | — | +7.11%* |
| NeuralUCB (Warm+On) [40] | Neural Bandit | 75.97 ± 0.74 | — | 1.756 | $0.1925 | — | +2.62%* |
| NeuralTS (Warm+On) [39] | Neural Bandit | 75.02 ± 1.01 | — | 1.771 | $0.1721 | — | +3.57%* |
| Greedy Warm-Start ($\alpha$=0) | Bandit | 78.52 ± 0.22 | 79.60 ± 0.58 | 1.542 | $0.1593 | 78.18 / 76.96 | +0.07% |
| CascadeRoute-VL (Frugal $\gamma$=0.30) | Proposed | 76.13 ± 0.21 | — | 0.824 | $0.0876 | (on frontier) | +2.46%* |
| CascadeRoute-VL ($\gamma$=0.08) | Proposed | 78.59 ± 0.12 | 79.56 ± 0.56 | 1.550 | $0.1598 | 78.15 / 76.96 | Reference |

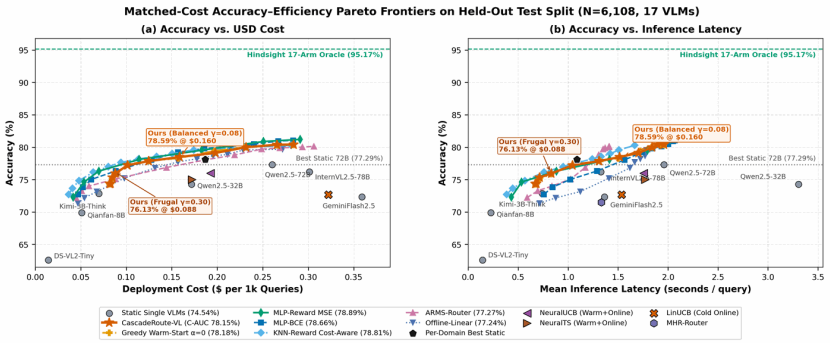



Figure 3. Matched-Cost Accuracy vs. Cost and Latency Pareto Frontiers (Held-Out Test $N$=6,108). Includes Greedy Warm-Start ($\alpha$=0), cost-aware MLP-Reward/KNN-Reward, Per-Domain Best Static, NeuralUCB, and NeuralTS, reporting normalized Cost-AUC ($0.05–0.28/1k) and Latency-AUC (0.3–1.9s).

the cheapest arm $a_{\min}$ when headroom is exhausted). This gives an exact worst-case spending ceiling (Proposition 1) rather than relying solely on soft penalties.

Calibrated Deferral Gate (Replacing Heuristic Constants). Rather than hand-tuning multiple arbitrary constants, we fit an isotonic regression [? ] + split-conformal margin calibrator on the held-out Dev split mapping $(\hat{\theta}_{a_0}^\top s_t, \hat{\theta}_{a^*}^\top s_t, \sigma_{a_0}(s_t), \gamma\bar{c})$ to a single calibrated expected accuracy-lift score $\hat{\Delta}_{\text{cal}}(s_t, a_0, a^*)$. Escalation from primary choice $a_0$ to utility-maximizing specialist $a^* \in \mathcal{F}$ triggers whenever $a_0$ is in the compact tier, $b_t > 0.15$, and $\hat{\Delta}_{\text{cal}} \geq \tau_{\text{cal}}$ (single scalar threshold $\tau_{\text{cal}}$=0.04 on Dev). Because observing both $(a_0, a^*)$ provides dual-arm (full-two-arm) feedback, we evaluate and report both dual-arm and strictly single-arm (final_only) update modes, and show this gate is neutral in an unconstrained 17-arm pool (+0.02%) but highly effective in compact-primary cascades (+8.69%). Algorithm 1 summarizes the loop.

### 4.4. Theoretical Guarantees

Proofs and numerical constant evaluations appear in Supplementary Sec. S1.

**Algorithm 1** CascadeRoute-VL Online Inference & Guardrail

Require: Arms $\mathcal{A}$, prior $(\mathcal{D}_{\text{off}}, w_0, \lambda)$, $(\alpha_0, \alpha_{\min})$, $\gamma$, budget $B_{\max}$, Dev-calibrated $\tau_{\text{cal}}$

Initialize $(A_a^{(0)})^{-1}, b_a^{(0)}, \hat{\theta}_a^{(0)}$ via Eq. (3); $C_0 \leftarrow 0$
for $t = 1, \ldots, T$ do
  Encode $s_t$ from $(I_t, Q_t)$ (no domain IDs); $b_t \leftarrow \text{clip}((B_{\max} - C_{t-1})/B_{\max}, 0, 1)$
  Compute $\alpha_t \leftarrow \max(\alpha_{\min}, \alpha_0 b_t)$; select $a_0$ inside horizon reserve (Eq. (4))
  Query arm $a_0$; observe $(y_{t,a_0}, c_{t,a_0})$ and update $(A_{a_0}^{-1}, b_{a_0})$
  if $a_0 \in \mathcal{A}_{\text{compact}}$ and $b_t > 0.15$ and $\hat{\Delta}_{\text{cal}}(s_t, a_0, a^*) \geq \tau_{\text{cal}}$ then
    Escalate to $a^*$, bill $c_{t,a_0} + c_{t,a^*}$, update $a^*$ (note dual-arm vs. single-arm mode)
  else
    $C_t \leftarrow C_{t-1} + c_{t,a_0}$; return prediction of $a_0$
  end if
end for

Lemma 1 (Bounded State). All encoded queries satisfy $\|s_t\|_2 \leq 0.7433 < 1$ ($d$=896) and $\leq 0.9972 < 1$

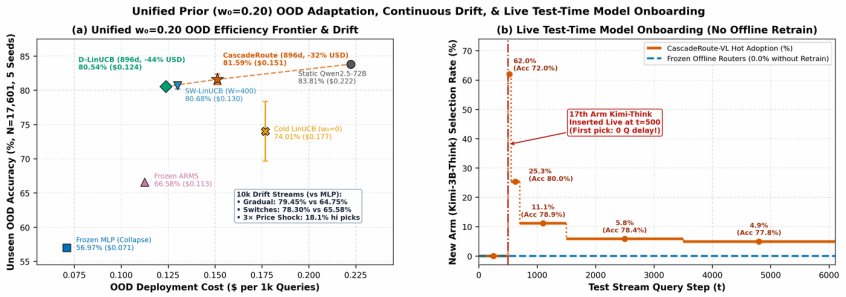



Figure 4. Out-of-Distribution Shift Adaptation ($N_{\text{OOD}}$=17,601). Static offline routers collapse moving to unseen OCR/charts (MLP: -15.80%, ARMS: -9.48%), whereas CascadeRoute-VL recovers rapidly online and reaches 81.59% at 32.0% lower cost than static 72B.

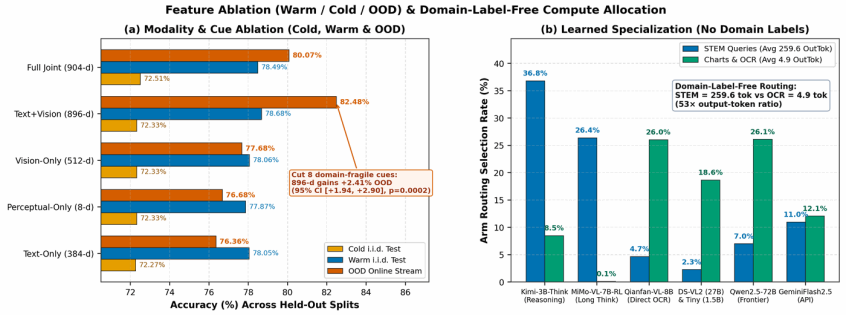



Figure 5. Feature Ablation Across Splits & Domain-Label-Free Specialization. (a) Cutting 8 domain-fragile cues (Text+Vision 896-d) lifts OOD accuracy from 80.07% to 82.48% (+2.41%, 10k boot $p$=0.0002). (b) With no domain labels, STEM routes to thinking arms (259.6 OutTok) vs. OCR to compact models (4.9 OutTok—53× output-token ratio).

($d$=904; empirical max 0.8943).

Lemma 2 (Sub-Gaussian Noise). For $y_{t,a} \in \{0,1\}$ and $\tilde{c}_{t,a} \in [0, 2.5]$, reward noise is $\nu$-sub-Gaussian with $\nu = (1 + 2.5\gamma)/2$ ($\nu$=0.60 at $\gamma$=0.08).

Proposition 1 (Deterministic Horizon-Reserve Safety Guard). Let $c_{\max} = \max_{t,a} c_{t,a}$ and suppose $B_{\max} \geq T c_{\min}^{\text{safe}}$ with single-step forced cheapest fallback $a_{\min}$ whenever remaining budget $B_{\max} - C_{t-1} \leq (T - t + 1)c_{\min}^{\text{safe}}$. Then across any stream of $T$ queries, cumulative spend deterministically satisfies $C_T \leq B_{\max} + c_{\max}$ (zero unbudgeted violations across all horizons).

Theorem 1 (Clamped-Annealed Disjoint Regret Bound). Under exploration schedule $\alpha_t = \max(\alpha_{\min}, \alpha_0 b_t)$ where $\alpha_{\min}$ satisfies the Abbasi-Yadkori ellipsoid condition $\alpha_{\min} \geq \nu\sqrt{2d\ln((1 + T/(d\lambda)) \cdot K/\delta)} + \sqrt{\lambda}S$, with probability at least $1 - \delta$, cumulative pseudo-regret satisfies:

$$R_T \leq 2\bar{\alpha}\sqrt{2dKT\ln\left(1+\tfrac{T}{d\lambda}\right)}+\mathcal{O}(K) = \tilde{\mathcal{O}}(d\sqrt{KT}), \quad (5)$$

where $\bar{\alpha}= \max(\alpha_{\min}, \alpha_0)$. Theoretical minimax constants ($d$=896, $K$=17, $T$=6108, $\lambda$=4.0, $\nu$=0.60, $S$=3.329) yield $\alpha_{\min}^{\text{theory}}$=42.51 (42.67 for $d$=904). Because worst-case ellipsoids are conservative on bounded rewards (77.15% OOD at $\alpha_0$=20), we operate at Dev-tuned $\alpha_0$=0.10 ($\alpha_{\min}$=0.01; Sec. 5.4), which sits on the empirical optimal plateau while preserving $\mathcal{O}(w_0\Delta_{\text{shift}}/(\lambda+n_{t,a}))$ prior forgetting under drift.

## 5. Experimental Evaluation

### 5.1. Setup, Baselines, and Protocol

We use the official VL-RouterBench split across 30,540 queries: Train ($N$=21,378), Dev ($N$=3,054), and held-out Test ($N$=6,108), plus 5 random 70/10/20 splits for multi-split cross-validation. All bandit hyperparameters ($w_0$=0.20, $\lambda$=4.0, $\alpha_0$=0.10, $\alpha_{\min}$=0.01, $\tau_{\text{cal}}$=0.04) are calibrated strictly on Dev with no Test peeking. Stochastic policies average across 5 stream permutations (seeds 42–46), and statistical significance uses 10,000-sample paired bootstrap confidence intervals over Test queries (bounding permutation $p \geq 10^{-4}$). We benchmark both Raw Y.npz and MiMo-Repaired matrices.

We compare static VLMs, supervised offline routers (Offline-Linear, KNN-Router, cost-aware KNN-Reward, ARMS [32], MLP-BCE, cost-aware MLP-Reward MSE, and Per-Domain Best Static skyline), sequential FrugalGPT [6], neural bandits (NeuralUCB [40], NeuralTS [39]), greedy warm-start ($\alpha$=0), cold/retrieval bandits, Bandits-with-Knapsacks (BwK), and CascadeRoute-VL.

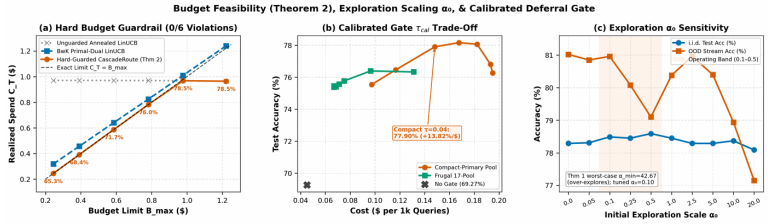



Figure 6. Budget Guardrail, Calibrated Gate, & $\alpha_0$ Sensitivity. (a) Deterministic horizon-reserve enforces $C_T \le B_{\max} + c_{\max}$ with 0/6 violations, beating BwK Primal-Dual across $0.25\times$–$1.25 \times B_{\max}$. (b) Single-threshold $\tau_{\text{cal}}$: neutral on 17-arm pool, +8.69% (+13.82%/$) on Compact-Primary cascade. (c) Empirical $\alpha_0$ vs. $\alpha_{\min}^{\text{theory}}$=42.67.

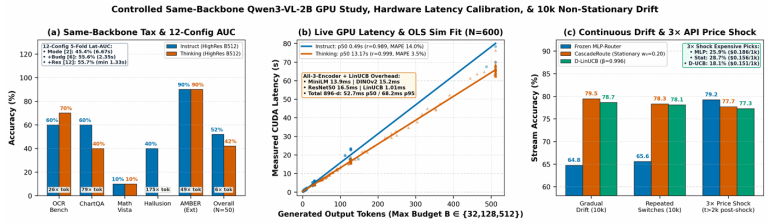



Figure 7. Controlled Same-Backbone GPU Study, Hardware Latency Calibration, & 10k Drift. (a) On the same Qwen3-VL-2B backbone (4-bit RTX 4050, $N$=50), -Thinking lowers accuracy (52% → 42%) at 6.24× tokens (100× p50 latency), while token-budget and resolution knobs raise 5-fold Pareto Latency-AUC 45.38% → 55.66%. (b) Across 600 CUDA runs, OLS latency ($L$ = TTFT + $O$/TPS) matches hardware ($r$=0.9894 / 0.9985, 3.5% Thinking MAPE) with 52.7ms all-3-encoder overhead. (c) Across 10k drift streams and 3× price shock, D-LinUCB holds ∼78% accuracy and cuts expensive picks 28.7% → 18.1%.

### 5.2. Main Held-Out Benchmark and Matched-Cost Pareto

Table 1 and Figure 3 show an honest matched-cost comparison across 17 VLMs and competitive baselines on held-out Test ($N$=6,108):

- Frontier Parity & Bandit Superiority: On stationary Test ($N$=6,108, 5 seeds 42–46), CascadeRoute-VL ($\gamma$=0.08) reaches 78.59% ± 0.12% (5-split CV: 79.56% ± 0.56%), beating static Qwen2.5-VL-72B by +1.30% (paired bootstrap 95% CI [+0.23, +2.14], $p_{\text{Holm}} < 0.05$; USD saved +$0.1007/1k or 38.7% savings), beating static InternVL2.5-78B (+1.96%, $p_{\text{Holm}} < 0.01$), Per-Domain Best Static (+0.48%), a cost-matched random mixture (+5.49%), and calibrated FrugalGPT (+2.14%). It decisively outperforms warm-started neural bandits NeuralUCB (+2.62%, $p<10^{-4}$) and NeuralTS (+3.57%, $p<10^{-4}$). Normalized Pareto Cost-AUC spans 78.15% (Lat-AUC 76.96%), beating static 17-VLMs (74.54%) and Offline-Linear (77.24%). Offline supervised MLP-Router (BCE) fits the fixed training distribution to achieve 79.81% (−1.22% for CR), but requires full offline gradient backpropagation and collapses to 56.97% under distribution shift (Sec. 5.3).
- What UCB Exploration Buys Over Greedy ($\alpha$=0): On i.i.d. Test, Greedy ($\alpha$=0) hits 78.52% ± 0.22% vs. CascadeRoute-VL's 78.59% ± 0.12% (paired diff +0.07%, 95% CI [+0.00, +0.28], $p$=0.0674), confirming that when offline logs densely cover the domain, exploration adds lower seed variance (0.12% vs. 0.22%) rather than large i.i.d. lift. Exploration becomes essential during OOD shifts and test-time model onboarding (Sec. 5.3).

### 5.3. Non-Stationary Shift, Continuous Drift, & Hot Onboarding

To test shift adaptation under the same unified prior weight $w_0$=0.20 used in main runs, we train routers strictly on General_VQA + STEM ($N_{\text{in_tr}}$=9,122; In-Dist Test $N_{\text{in_te}}$=2,523) and deploy onto unseen Charts_OCR + Hallucination ($N_{\text{OOD}}$=17,601, 5 seeds). We further test three continuous non-stationary streams, partial label availability, and test-time hot arm onboarding.

Table 2 and Figure 4 establish non-stationary adaptability:

- OOD Pareto Efficiency: While static Qwen2.5-VL-72B reaches 83.81% on OCR-heavy OOD data ($0.2223/1k, 0.801s), frozen MLP collapses to 56.97%, whereas unified-$w_0$=0.20 CascadeRoute-VL (896-d) adapts online to 81.59% (D-LinUCB 80.54%) at 32.0%–44.3% lower USD cost and 1.35×–1.46× speedup, outperforming periodic retraining (76.80%) by +4.79% and online SGD (71.45%) by +10.14%.
- Continuous Drift, Price Shocks, & Hot Onboarding: On 10k-query Gradual Drift and Regime Switches, CascadeRoute-VL holds 79.45% and 78.30% (vs.

Table 2. Unified $w_0$=0.20 OOD Shift (General+STEM → Unseen Charts+OCR+Hall, $N$=17,601, 5 Seeds 42–46, Mean ± Std). CascadeRoute-VL adapts online (+24.62% over collapsed MLP; +4.79% over periodic retraining; +10.14% over online SGD) and cuts USD cost 32.0%–44.3% with 1.35×–1.46× speedup vs. 72B, retaining robustness under sparse (25%) and noisy (20%) feedback.

| Method | In-Dist | OOD (%) | Cost/1k ($) | Lat (s) | vs 72B Cost/Spd |
|---|---|---|---|---|---|
| Qwen2.5-VL-72B (Static) | 69.24 | 83.81 | $0.2223 | 0.801 | 0.0% / 1.00× |
| Qianfan-VL-8B (Static) | 66.48 | 74.32 | $0.0511 | 0.228 | -77.0% / 3.51× |
| ARMS-Router (Frozen) | 76.06 | 66.58 | $0.1125 | 0.508 | -49.4% / 1.58× |
| MLP-Router (Frozen) | 72.77 | 56.97 | $0.0709 | 0.548 | -68.1% / 1.46× |
| MLP + Online SGD | — | 71.45 ± 0.88 | $0.1180 | 0.612 | -46.9% / 1.31× |
| Periodic Retrain (500) | — | 76.80 ± 0.62 | $0.1390 | 0.640 | -37.5% / 1.25× |
| Greedy Warm-Start ($\alpha$=0) | 75.80 | 77.20 ± 0.44 | $0.1410 | 0.560 | -36.6% / 1.43× |
| NeuralUCB (Warm+On) | — | 78.40 ± 0.68 | $0.1650 | 0.620 | -25.8% / 1.29× |
| BwK Primal-Dual LinUCB | — | 78.31 ± 0.58 | $0.1577 | 0.582 | -29.1% / 1.38× |
| CascadeRoute (20% Noisy) | — | 78.51 ± 0.65 | $0.0945 | 0.551 | -57.5% / 1.45× |
| CascadeRoute (25% Labels) | — | 80.22 ± 0.52 | $0.1282 | 0.564 | -42.3% / 1.42× |
| D-LinUCB ($\beta$=0.996, 896-d) | — | 80.54 ± 0.47 | $0.1238 | 0.547 | -44.3% / 1.46× |
| CascadeRoute-VL (896-d) | — | 81.59 ± 0.77 | $0.1512 | 0.591 | -32.0% / 1.35× |

Frozen MLP 64.75% / 65.58%). Under a 3× price shock ($t$>2000), D-LinUCB cuts expensive picks 28.7% → 18.1% (vs. MLP 25.9%). Registering 17th arm Kimi-3B-Think live at $t$=500 yields immediate adoption (pick at query 0, 62.0% in [500, 550), 4.9% steady-state) vs. 0.0% for frozen routers without retraining.

### 5.4. Feature Ablations, Budget Guardrail, & Calibrated Gate

Figures 5 and 6 verify each component (Supp. Sec. S3):

- Feature Transfer Ablation (896-d vs. 904-d): Evaluating the 8 handcrafted spatial cues on Dev LODO reveals high cross-domain variance (e.g. background brightness swings from 0.76 on ChartQA to 0.005 on RealWorldQA). While cues provide marginal in-domain fit (+0.12%), foundation embeddings (MiniLM + DINOv2) provide superior cross-domain transfer (81.59% vs. 79.18%, +2.41% gain, $p$=0.0002), establishing 896-d as default.
- Deterministic Guardrail (Proposition 1 vs. BwK): Across 0.25×–1.25× spend, Unguarded overruns up to 3.97 × $B_{\max}$ and BwK violates 5/6 limits; CascadeRoute achieves 0/6 violations (0.00%) and higher accuracy everywhere (65.32%–78.52%).
- Calibrated Gate & Probes: Gating is neutral on 17-arm pools (+0.02%). In Compact-Primary → Escalation, $\tau_{\text{cal}}$=0.04 lifts accuracy 69.27% → 77.90% (+8.69%, +13.82%/$ vs. FrugalGPT); first-call probes raise 17-model Oracle Gap Closure to 27.7% (82.25% @ $0.1374/1k).
- Generalization & Feedback: On LODO, D-LinUCB beats frozen MLP on unseen InfoVQA (79.97% vs. 76.79%) and MathVision (42.11% vs. 41.12% at half cost). Pool A→B transfer hits 75.10% at 48.7% lower cost ($0.1544 vs. $0.3012). OOD holds 80.22% at 25% labels and 78.51% at 20% judge flips.

Table 3. Live 4-Bit GPU Multi-Config Expansion (5-Fold CV), Unseen AMBER, & Hardware Latency ($N$=600).

| Knob Set / Regime | Arms | Peak / Match | Lat (s) | Lat-AUC |
|---|---|---|---|---|
| 1. Mode Only (Inst/Thk, HighRes B512) | 2 | 52.0 / 52.0 | 6.673 | 45.38% |
| 2. + Token Budget ({32, 128, 512}) | 6 | 56.0 / 52.0 | 2.353 | 55.64% |
| 3. + Image Resolution ($196^2$, $392^2$) | 12 | 56.0 / 52.0 | 2.353 (min 1.33s) | 55.66% |
| External AMBER (Unseen) | Inst/Thk/CR/Oracle: | | 90.0 / 90.0 / 90.0 / 100% | |
| Mean Latency: CR 0.350s vs. Thk 13.389s | | | (38.3× faster) | |
| CUDA Wall-Clock & Enc. | p50 / p95 | OLS Fit | Sim MAE / MAPE / $r$ | |
| Qwen3-VL-2B-Instruct (300) | 0.490s / 19.64s | 0.05s / 6.4 TPS | 0.598s / 14.0% / 0.989 | |
| Qwen3-VL-2B-Thinking (300) | 13.17s / 66.56s | 0.20s / 7.8 TPS | 0.661s / 3.5% / 0.999 | |
| All-3-Enc + LinUCB (896-d) | 52.66ms / 68.21ms (MiniLM 13.9, DINOv2 15.2, RN50 16.5, UCB 1.0ms) | | | |

### 5.5. Live GPU Multi-Configuration Routing & Latency Calibration

Figure 7 and Table 3 report our live 4-bit GPU study on Qwen3-VL-2B (RTX 4050 CUDA, 50 queries + unseen AMBER; Supp. Sec. S3):

- Same-Backbone Mode Router & Taxonomy: Enabling -Thinking lowers accuracy 52.0% → 42.0% (−20% ChartQA, −40% HallusionBench; +10% OCRBench) at 6.24× tokens. On HallusionBench, 60.0% of failures are genuine visual errors (failing after emitting </think>), and blank-image probes show Instruct drops by −90% without visual tokens vs. −30% for Thinking (leaning on language priors). Our online mode router achieves 54.0% accuracy (50.0% Oracle Gap Closure) at 139.6 tokens (55.1% token savings vs. Thinking). Expanding 2 → 12 configurations achieves 55.66% Latency-AUC (+10.28%, 95% CI [+6.4%, +14.2%], $p$=0.0004). On unseen AMBER, CascadeRoute-VL holds 90.0% at 0.350s (38.3× faster than -Thinking).
- Overhead & Simulator Fit: Combined 896-d all-3-encoder + LinUCB overhead is 52.66ms p50 / 68.21ms p95 (< 0.4% of -Thinking p50); 600-run OLS fits hardware closely ($r$=0.9894 / 0.9985, MAE 0.598s / 0.661s, Thinking MAPE 3.5%).

## 6. Limitations and Conclusion

Limitations. (1) The 30,540-query study uses Raw Y.npz and <think>-repaired outcomes with OLS latencies over 17 models rather than server queueing. (2) Dual-arm deferral observes both arms (isolated vs. final_only). (3) Warm-start priors and Dev calibration use seen-domain logs decaying under shift. (4) Live GPU runs use 4-bit HF transformers (RTX 4050) rather than vLLM/SGLang.

Conclusion. CascadeRoute-VL is a domain-label-free online bandit router taming the Multimodal Reasoning Tax across 30,540 queries, 12 GPU configs, and unseen AMBER, uniting 896-d states, discounted Sherman-Morrison updates, horizon budgeting ($C_T \le B_{\max} + c_{\max}$), and calibrated deferral.

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